

Graph-Aware Safe Volt–VAR Control for PV-Rich Distribution Feeders

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ABSTRACT

Distribution feeders with high photovoltaic (PV) penetration experience fast and time-varying voltage changes, while inverter reactive-power commands can propagate through the network and influence nearby buses. This paper develops a graph/neighborhood-aware safe Volt–VAR controller for inverter-interfaced radial distribution feeders under time-varying PV and load disturbances. The proposed controller uses local voltage feedback together with limited one-hop controllable-neighbor information to construct a graph-based nominal reactive-power command. Before dispatch, the nominal command is passed through a stability-oriented action filter that applies conservative gain scaling, inverter saturation limits, and ramp-rate constraints. This structure accounts for nearby voltage interactions while avoiding the computational burden of centralized optimal power flow or online actor–critic deployment. The method is evaluated on a single-phase IEEE-13-inspired radial feeder with controllable inverters at selected buses. Compared with the reported baseline, the proposed controller reduces the mean recovery time from 2.6 s to 1.0 s and improves the mean transient reward from -6.76 to -1.50 , while maintaining a small mean stability index of 1.5×10^{-4} . The results show that limited neighborhood information and conservative safety filtering can improve transient voltage regulation while keeping inverter commands bounded for real-time deployment.

Keywords: Volt–VAR control, smart inverters, photovoltaic generation, voltage regulation, graph-based control, safe control.

1. Introduction

The rapid integration of photovoltaic (PV) generation into active distribution networks has significantly changed the operating characteristics of modern power systems. Unlike conventional bulk generation, PV resources are often installed close to load centers and are strongly affected by local irradiance, cloud movement, and customer-side operating conditions. As a result, high PV penetration can introduce rapid active-power variations, reverse power flow, voltage rise, and increased difficulty in maintaining feeder voltages within acceptable operating limits. NRE (2025); Turitsyn, Šulc, Backhaus and Chertkov (2011); Ste (2012); Vol (2025). These challenges are further complicated by stochastic load demand, distribution-network reconfiguration, and the limited observability normally available in practical distribution systems ADN (2024); DN (2024); DR (2023). Therefore, real-time voltage regulation in PV-rich feeders requires control strategies that are fast, scalable, and capable of accounting for the coupling between electrically neighboring buses.

Smart inverters provide an important opportunity for addressing this problem. Modern interconnection standards and grid-support guidelines allow inverter-interfaced distributed energy resources (DERs) to participate in reactive-power support, Volt–VAR regulation, ride-through operation, and other grid-support functions IEE (2019, 2020, 2018). Autonomous smart-inverter Volt–VAR functions have been studied as practical tools for voltage regulation, voltage reduction, and power-quality improvement in distribution feeders Ding, Nguyen, Walinga, Nagarajan, Baggu, Chakraborty, McCarty and Bell (2017); Sup (2024). In addition to inverter control, other supporting technologies such as battery energy storage systems, fast frequency response services, synthetic inertia, and IoT-based monitoring have been investigated to improve the resilience and observability of PV-rich grids FFR (2023); Rap (2019); ESS (2023);

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AEM (2024); IoT (2025, 2024); Adv (2025); Rea (2024); AI (2025). However, these technologies do not remove the need for local inverter-level voltage controllers that can operate quickly under time-varying PV and load disturbances.

A central difficulty in inverter-based Volt-VAR control is that voltage regulation is not purely local. Due to line impedance and the radial structure of distribution feeders, reactive-power injection at one bus can influence not only the local bus voltage but also the voltages of electrically nearby buses. Classical and distributed reactive-power feedback methods have shown that network sensitivity and communication structure play key roles in distribution voltage regulation Turitsyn et al. (2011); Dis (2015); Bolognani, Carli, Cavraro and Zampieri (2019). Centralized or optimal-power-flow-based approaches can coordinate multiple devices, but they often require broad feeder observability, reliable communication, and repeated optimization, which can limit their use in fast real-time inverter dispatch OPF (2016); Dis (2013); Sma (2022). On the other hand, purely local controllers are simpler to implement but may create conflicting actions among nearby inverters if feeder coupling is not properly considered.

Recent research has explored several directions for improving voltage control in active distribution networks. Forecasting-based approaches, including machine-learning models for short-term PV prediction, can improve the estimation of future PV output and support better dispatch decisions Lig (2023a,b); Sol (2025). However, forecasting methods are usually complementary to the control layer and do not by themselves guarantee closed-loop voltage stability. Multi-agent deep reinforcement learning has also been studied for decentralized voltage control, including methods that coordinate active and reactive power and methods that incorporate power-system structure into the learning process MAD (2022); Phy (2023). Attention-enabled and transformer-based multi-agent controllers further attempt to reduce unnecessary communication by allowing agents to focus on the most relevant neighboring information Cao, Zhao, Hu, Ding, Huang and Chen (2021); Tra (2022). These approaches are promising, but their deployment can be challenging when training complexity, stability certification, communication reliability, and real-time implementation constraints must be considered simultaneously.

Other studies have focused on topology adaptation, network reconfiguration, and multi-timescale coordination. Topology-aware voltage-regulation methods and reconfiguration-aware control strategies are important because distribution feeders may change structure during switching events, outages, or operational reconfiguration Top (2022); DR (2023). Meta-transfer learning and related adaptation methods have also been explored for power-system fault analysis and topology-varying operating conditions Met (2024). In parallel, two-timescale and multi-timescale voltage-control frameworks have been proposed to coordinate fast inverter actions with slower legacy devices or utility equipment Two (2022, 2023); Mul (2022). These studies highlight the importance of adaptability and coordination, but they also introduce additional modeling, training, and implementation complexity.

Stability assurance is another major concern for learning-based and decentralized voltage controllers. Safe reinforcement learning, Lyapunov-based control, and stability-constrained reinforcement learning have been proposed to reduce unsafe exploration and prevent destabilizing actions Saf (2025); Neu (2019); Lya (2024, 2022); Off (2025). For decentralized real-time voltage control, stability-constrained methods show that the feedback gain and the voltage-sensitivity matrix must be considered together to avoid unstable closed-loop behavior Feng, Shi, Qu, Low, Anandkumar and Wierman (2024). These results motivate controller designs that do not rely only on nominal performance, but also include explicit safety mechanisms such as gain limits, saturation constraints, and ramp-rate constraints.

Despite this progress, there remains a practical gap between advanced coordination methods and lightweight real-time inverter deployment. Fully centralized methods may require extensive observability and optimization, whereas unconstrained learning-based controllers may require careful training and may produce unsafe actions without additional filtering. At the same time, purely local Volt-VAR controllers may neglect neighbor-bus interactions caused by feeder sensitivity. This motivates a controller that is simple enough for real-time inverter implementation, but still accounts for local network coupling and incorporates a stability-oriented safety mechanism.

To address this gap, this paper develops a graph/neighborhood-aware safe Volt-VAR controller for inverter-interfaced radial distribution feeders. The proposed controller uses local voltage feedback together with limited one-hop controllable-neighbor information to construct a graph-based nominal reactive-power command. The nominal command is then passed through a stability-oriented action filter that applies conservative gain scaling, inverter command saturation, and ramp-rate limits before dispatch. In contrast to a full centralized optimal-power-flow controller, the proposed method does not require complete feeder-wide optimization at every control step. In contrast to an online actor-critic deployment pipeline, the implemented controller uses an interpretable graph-based feedback structure with an explicit safety filter.

The main contributions of this work are summarized as follows:

Table 1

Notation for the Linearized DistFlow Model

Symbol	Description
$\mathcal{G} = (\mathcal{N}_0, \mathcal{E})$	Radial distribution feeder graph.
$\mathcal{N}_0 = \{0, 1, \dots, n\}$	Set of all buses, including the slack bus.
$\mathcal{N} = \{1, \dots, n\}$	Set of non-slack buses.
$\mathbf{v} \in \mathbb{R}^n$	Vector of squared voltage magnitudes at buses $1, \dots, n$.
$\mathbf{q} \in \mathbb{R}^n$	Reactive-power injections from inverter-interfaced DERs.
$\mathbf{u} \in \mathbb{R}^n$	Reactive-power control input or ramp-rate command.
$\mathbf{X} \in \mathbb{R}^{n \times n}$	Reactance sensitivity matrix relating reactive-power injections to voltage changes.
$\mathbf{v}^{\text{env}}$	Uncontrolled voltage component due to load demand, PV active power, and the slack bus.

- **Graph/neighborhood-aware Volt-VAR formulation:** A decentralized inverter-control structure is developed in which each controllable bus uses its local voltage deviation and limited one-hop controllable-neighbor information to account for feeder coupling and reduce conflicting reactive-power actions.
- **Stability-oriented safety filtering:** A practical action-filtering mechanism is introduced to conservatively scale the nominal control command and enforce inverter saturation and ramp-rate limits. The design is motivated by stability-constrained voltage-control principles and is suitable for real-time implementation.
- **Lightweight implementation without online centralized optimization:** The controller avoids solving a full centralized optimal-power-flow problem or a quadratic safety projection at every time step, reducing computational complexity while retaining voltage-coupling awareness.
- **Dynamic case-study validation:** The controller is evaluated on a single-phase IEEE-13-inspired radial feeder under time-varying PV and load disturbances. The results show faster mean recovery time and improved transient reward compared with the reported baseline, while maintaining bounded inverter command behavior.

The remainder of the paper is organized as follows. Section II introduces the distribution feeder model and formulates the Volt-VAR control problem. Section III presents the graph/neighborhood-aware safe controller and the stability-oriented action filter. Section IV describes the experimental setup, including the test feeder, disturbance profiles, controller implementation, and performance metrics. Section V presents the case-study results, and Section VI concludes the paper with limitations and future research directions.

2. System Model and Problem Formulation

We consider a radial distribution feeder represented by a graph $\mathcal{G} = (\mathcal{N}_0, \mathcal{E})$, where $\mathcal{N}_0 = \{0, 1, \dots, n\}$ denotes the set of buses, bus 0 is the substation or slack bus, and $\mathcal{E}$ is the set of distribution lines. The set of non-slack buses is denoted by $\mathcal{N} = \{1, \dots, n\}$. A subset of buses is equipped with inverter-interfaced distributed energy resources (DERs) capable of providing reactive-power support for voltage regulation. Such inverter-based Volt-VAR control is widely used for distribution networks with high photovoltaic (PV) penetration Turitsyn et al. (2011); Ding et al. (2017); IEE (2020).

Under the standard assumptions of small voltage deviations around a nominal operating point, radial feeder structure, and a linearized branch-flow approximation, the squared voltage magnitudes can be expressed as an affine function of reactive-power injections Dis (2015); Bolognani et al. (2019); Feng et al. (2024). Specifically, the voltage profile is modeled as

$$\mathbf{v} = \mathbf{X}\mathbf{q} + \mathbf{v}^{\text{env}}, \quad (1)$$

where $\mathbf{v} \in \mathbb{R}^n$ is the vector of squared voltage magnitudes at the non-slack buses, $\mathbf{q} \in \mathbb{R}^n$ is the reactive-power injection vector, $\mathbf{X} \in \mathbb{R}^{n \times n}$ is the reactance sensitivity matrix, and $\mathbf{v}^{\text{env}}$ represents the uncontrolled voltage component caused

by the slack-bus voltage, active-power PV injections and load demand. The notation used in this model is summarized in Table 1.

The matrix $\mathbf{X}$ captures the electrical coupling among buses. As a result, a reactive-power action at one inverter can affect not only the local bus voltage but also electrically nearby buses. This coupling is one of the main reasons why purely local inverter actions may lead to undesirable neighbor-bus voltage deviations or oscillatory behavior when control actions are aggressive Turitsyn et al. (2011); Bolognani et al. (2019).

Each inverter updates its reactive-power command at discrete time instants $t = 0, 1, 2, \dots$ with a fixed sampling interval $\Delta T > 0$.

The incremental reactive-power dynamics are modeled as

$$\mathbf{q}(t+1) = \mathbf{q}(t) + \Delta T \mathbf{u}(t), \quad (2)$$

where $\mathbf{u}(t)$ denotes the vector of reactive-power ramp-rate commands.

$$\begin{aligned} \mathbf{v}(t+1) &= \mathbf{X}\mathbf{q}(t+1) + \mathbf{v}^{\text{env}} \\ &= \mathbf{X}(\mathbf{q}(t) + \Delta T \mathbf{u}(t)) + \mathbf{v}^{\text{env}} \\ &= \mathbf{v}(t) + \Delta T \mathbf{X}\mathbf{u}(t). \end{aligned} \quad (3)$$

Thus, the discrete-time voltage dynamics are written as

$$\mathbf{v}(t+1) = \mathbf{v}(t) + \Delta T \mathbf{X}\mathbf{u}(t). \quad (4)$$

Equation (4) shows that the voltage response is directly shaped by the sensitivity matrix $\mathbf{X}$. Therefore, the control problem is not only to regulate each local voltage but also to account for the network-induced coupling between neighboring buses. This motivates the use of a neighborhood-aware controller together with a stability-oriented safety mechanism.

The objective of the inverter controller is to select the command $\mathbf{u}(t)$ such that the resulting voltages remain within acceptable operating limits while avoiding excessive reactive-power movement. The controller is designed to satisfy three requirements:

1. maintain bus voltages within prescribed admissible limits;
2. reduce neighbor-bus voltage interactions caused by the coupling matrix $\mathbf{X}$;
3. respect inverter command limits and ramp-rate constraints.

2.0.1. Voltage and Control Constraints

Each monitored bus voltage is required to satisfy

$$\underline{v} \leq v_i(t) \leq \bar{v}, \quad \forall i \in \mathcal{N}, \forall t, \quad (5)$$

where $\underline{v}$ and $\bar{v}$ denote the lower and upper voltage limits, respectively. Maintaining voltage within admissible limits is a standard requirement in distribution-system operation and smart-inverter Volt-VAR control IEE (2018, 2020); Ding et al. (2017).

The inverter control command is constrained by

$$\underline{u} \leq u_i(t) \leq \bar{u}, \quad \forall i \in \mathcal{N}, \forall t, \quad (6)$$

where $\underline{u}$ and $\bar{u}$ are the lower and upper allowable control limits. To avoid abrupt changes in reactive-power commands, a ramp-rate constraint is also imposed:

$$|u_i(t) - u_i(t-1)| \leq \Delta u_{\max}, \quad \forall i \in \mathcal{N}, \forall t. \quad (7)$$

These constraints reflect practical inverter operating limits and help reduce oscillatory control behavior during fast voltage transients Turitsyn et al. (2011); Sup (2024); Feng et al. (2024).

Overall Framework of the Proposed Graph-Aware Safe Volt-VAR Control

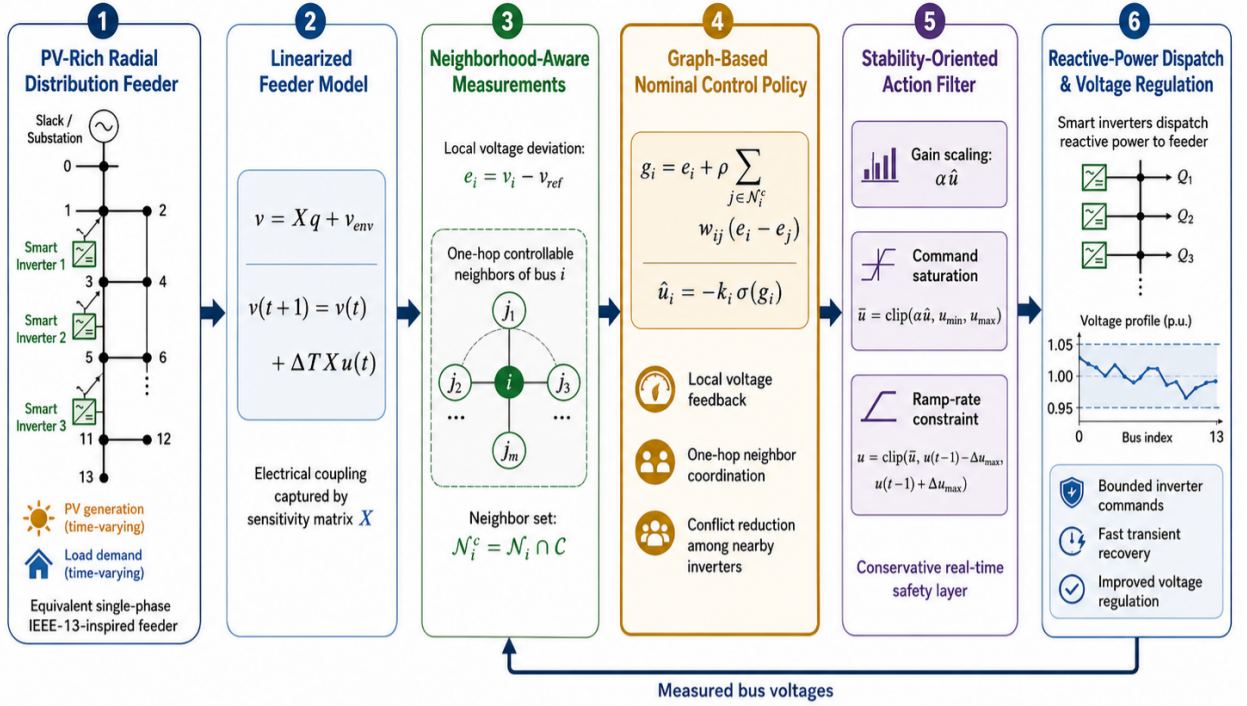


Fig. 1: Graph-Aware Safe Volt-VAR Control Scheme

3. Controller Architecture

This section presents the neighborhood-aware safe control architecture used for real-time inverter-based Volt-VAR regulation. The controller is designed around two components: first, a graph/neighborhood-aware nominal policy that computes reactive-power commands from local and neighboring voltage measurements; second, a stability-oriented action filter that enforces practical inverter limits and conservatively restricts the control gain to avoid destabilizing voltage responses. The design follows the stability-constrained voltage-control perspective developed for decentralized inverter control in distribution networks Dis (2015); Bolognani et al. (2019); Feng et al. (2024).

Let $\mathcal{C} \subseteq \mathcal{N}$ denote the set of controllable buses equipped with inverter-interfaced DERs. For each controllable bus $i \in \mathcal{C}$, the controller observes the local voltage and a limited neighborhood of electrically adjacent buses. The final dispatched command is obtained from a nominal graph-based command followed by a safety filtering step:

$$\mathbf{u}(t) = \mathcal{S}(\hat{\mathbf{u}}(t), \mathbf{u}(t-1), \mathbf{X}), \quad (8)$$

where $\hat{\mathbf{u}}(t)$ is the nominal command generated by the neighborhood-aware policy and $\mathcal{S}(\cdot)$ denotes the stability-oriented filter. Unlike an unconstrained learning policy, the filtered action is explicitly bounded by inverter command limits and ramp-rate limits before dispatch.

3.1. Stability Motivation

Under the linearized voltage dynamics in (4), the control law directly determines the evolution of bus voltages through the sensitivity matrix $\mathbf{X}$. Therefore, overly aggressive local reactive-power actions can propagate through the feeder and produce neighbor-bus voltage deviations or oscillatory behavior. A sufficient small-signal stability condition for decentralized voltage control of the form $\mathbf{u} = \pi(\mathbf{v})$ is Feng et al. (2024)

$$-\frac{2}{\Delta T} \mathbf{X}^{-1} < \frac{\partial \mathbf{u}}{\partial \mathbf{v}} < 0. \quad (9)$$

The right-hand inequality imposes negative voltage feedback, meaning that an increase in voltage should not produce an increase in reactive-power injection. The left-hand inequality limits the magnitude of the feedback gain so that the closed-loop voltage response remains stable under the discrete-time dynamics. This condition motivates the use of monotone voltage feedback together with a conservative gain-scaling mechanism.

In the implemented controller, the stability condition is not enforced by solving a full quadratic projection problem at every time step. Instead, the nominal graph-based command is passed through a conservative safety filter that scales and clips the command according to the stability margin, inverter limits, and ramp-rate limits. This provides a practical real-time approximation of stability-constrained voltage control while avoiding the computational burden of online optimization.

3.2. Neighborhood-Aware State Representation

For each controllable bus $i \in \mathcal{C}$, the local voltage deviation is defined as

$$e_i(t) = v_i(t) - v_{\text{ref}}, \quad (10)$$

where v_{ref} is the desired nominal squared-voltage reference. The neighborhood of bus i is denoted by $\mathcal{N}_i$, and the set of controllable one-hop neighbors is denoted by

$$\mathcal{N}_i^c = \mathcal{N}_i \cap \mathcal{C}. \quad (11)$$

The controller uses the local voltage deviation together with the voltage deviations of controllable neighboring buses:

$$\mathbf{s}_i(t) = \begin{bmatrix} e_i(t), \{e_j(t)\}_{j \in \mathcal{N}_i^c} \end{bmatrix}. \quad (12)$$

This representation preserves the decentralized nature of the controller while allowing each inverter to account for nearby voltage interactions caused by feeder coupling. Such communication-aware or neighborhood-aware voltage-control structures are important in distribution systems because reactive-power actions at one bus can affect voltages at other buses through the network sensitivity matrix Turitsyn et al. (2011); Bolognani et al. (2019).

3.3. Graph-Based Nominal Control Policy

The nominal controller computes a reactive-power command from a graph-based voltage feature. For bus i , a neighborhood potential is defined as

$$g_i(t) = e_i(t) + \rho \sum_{j \in \mathcal{N}_i^c} w_{ij} (e_i(t) - e_j(t)), \quad (13)$$

where $w_{ij} \geq 0$ is the electrical-neighborhood weight between buses i and j , and $\rho \geq 0$ controls the strength of neighbor coupling. The first term drives local voltage regulation, while the second term penalizes disagreement between neighboring controllable buses. This term helps reduce conflicting actions among nearby inverters.

The nominal reactive-power ramp command is then generated as

$$\hat{u}_i(t) = -k_i \sigma(g_i(t)), \quad (14)$$

where $k_i > 0$ is the local feedback gain and $\sigma(\cdot)$ is a bounded monotone nonlinearity. The negative sign enforces the desired Volt-VAR feedback direction: when the voltage deviation is positive, the inverter command is driven in the direction that reduces over-voltage; when the voltage deviation is negative, the command is driven in the opposite direction to support voltage recovery. This is consistent with standard inverter-based reactive-power voltage-control behavior Turitsyn et al. (2011); Ding et al. (2017); IEE (2020).

The monotonicity used in this work is imposed with respect to the local voltage feedback direction. Therefore, the intended structural behavior is

$$\frac{\partial \hat{u}_i}{\partial v_i} \leq 0, \quad (15)$$

rather than a stronger claim that every neighbor partial derivative is non-positive. Neighboring voltages enter the graph feature in (13) to coordinate nearby inverter behavior, but the primary monotone feedback guarantee is associated with the local voltage response.

3.4. Stability-Oriented Action Filter

The nominal command $\hat{\mathbf{u}}(t)$ is not dispatched directly. Instead, it is passed through a safety filter that applies conservative gain scaling, box constraints, and ramp-rate constraints. The filtered command is written as

$$\mathbf{u}(t) = \Pi_{\mathcal{U}}(\alpha \hat{\mathbf{u}}(t)), \quad (16)$$

where $\alpha \in (0, 1]$ is a conservative stability-scaling factor and $\Pi_{\mathcal{U}}(\cdot)$ denotes projection onto the practical inverter constraint set

$$\mathcal{U} = \{ \mathbf{u} : \underline{\mathbf{u}} \leq \mathbf{u} \leq \bar{\mathbf{u}}, \\ |\mathbf{u} - \mathbf{u}(t-1)| \leq \Delta \mathbf{u}_{\max} \}. \quad (17)$$

The scaling factor α is selected conservatively using the feeder sensitivity matrix and the feedback-gain bound suggested by (9). In this way, the safety layer limits the effective feedback gain before applying inverter saturation and ramp-rate constraints.

The action-filtering procedure can be summarized as

$$\tilde{\mathbf{u}}(t) = \alpha \hat{\mathbf{u}}(t), \quad (18)$$

$$\bar{\mathbf{u}}(t) = \text{clip}(\tilde{\mathbf{u}}(t), \underline{\mathbf{u}}, \bar{\mathbf{u}}), \quad (19)$$

$$\mathbf{u}(t) = \text{clip}(\bar{\mathbf{u}}(t), \mathbf{u}(t-1) - \Delta \mathbf{u}_{\max}, \mathbf{u}(t-1) + \Delta \mathbf{u}_{\max}). \quad (20)$$

Here, $\text{clip}(x, a, b)$ denotes element-wise clipping of x between the lower bound a and the upper bound b . Equation (18) limits the control gain, (19) enforces inverter command bounds, and (20) prevents abrupt changes in the dispatched reactive-power command. These operations are computationally lightweight and suitable for real-time control.

3.5. Relation to Reinforcement Learning-Based Controllers

Multi-agent reinforcement learning and attention-based coordination have been studied for decentralized voltage control in active distribution networks MAD (2022); Phy (2023); Cao et al. (2021); Tra (2022). In such methods, an actor network may be used to generate nominal inverter actions, while a critic or value function is used during training to improve long-term performance. However, unconstrained reinforcement learning policies may produce actions that violate inverter limits or destabilize fast voltage dynamics if stability is not explicitly considered Saf (2025); Lya (2024); Feng et al. (2024).

In this work, the implemented control architecture emphasizes a deterministic graph/neighborhood-aware safe controller rather than an online DDPG training pipeline. The graph-based nominal policy provides interpretable local feedback and limited neighbor coordination, while the stability-oriented filter enforces the practical constraints needed for real-time inverter dispatch. This choice reduces implementation complexity and avoids relying on a learned actor-critic model during deployment.

3.6. Controller Summary

The resulting controller operates as follows. At each sampling instant, each controllable inverter measures its local voltage and obtains the voltage measurements of its controllable one-hop neighbors. A graph-based voltage potential is computed from these measurements and used to generate a nominal reactive-power ramp command. The nominal command is then conservatively scaled using the feeder sensitivity information and filtered through inverter box and ramp-rate constraints before being dispatched.

The controller therefore combines three properties:

- **Local voltage feedback:** each inverter responds directly to its own voltage deviation;
- **Neighborhood coordination:** one-hop controllable neighbors are used to reduce conflicting local actions;
- **Safety filtering:** gain scaling, saturation, and ramp-rate constraints prevent excessive or abrupt reactive-power commands.

This architecture is consistent with stability-constrained decentralized voltage-control principles while remaining simple enough for real-time implementation on distribution feeders Dis (2015); Bolognani et al. (2019); Feng et al. (2024).

4. Experimental Setup

4.1. Test Feeder

The proposed controller is evaluated on a single-phase radial distribution feeder inspired by the IEEE 13-node benchmark. The feeder is represented by a slack bus and a set of downstream load and inverter-interfaced DER buses. The simulation uses an equivalent single-phase representation so that the voltage response can be modeled through the linearized sensitivity relation in (1). This representation allows the study to focus on the interaction between inverter reactive-power commands, feeder coupling, and voltage stability without modeling the full three-phase unbalanced feeder.

The feeder topology is radial, and the substation bus is treated as the fixed slack bus. Inverter-interfaced DERs are placed at a subset of controllable buses, denoted by $C \subseteq \mathcal{N}$. These inverters provide reactive-power support for voltage regulation, while the active-power variations from PV generation and load demand are treated as exogenous disturbances. Such an inverter-based Volt-VAR control setting is consistent with modern distribution-system operation under high PV penetration Turitsyn et al. (2011); Ding et al. (2017); IEE (2020).

The network coupling is captured by the reactance sensitivity matrix $\mathbf{X}$. Since $\mathbf{X}$ maps reactive-power injections to voltage changes, a command applied at one controllable bus can influence voltages at other electrically nearby buses. This makes the feeder suitable for evaluating neighbor-aware voltage-control behavior and the effect of stability-oriented action filtering. The importance of communication and network sensitivity in decentralized voltage regulation has been emphasized in prior work Dis (2015); Bolognani et al. (2019); Feng et al. (2024).

It should be noted that the feeder used in the simulation is an equivalent single-phase test system rather than a full three-phase unbalanced implementation of the IEEE 13-node benchmark. Therefore, the numerical results reported in this study should be interpreted as validating the proposed graph-based safe control mechanism under a representative radial feeder model, not as a complete three-phase feeder study.

4.2. Time-Varying PV and Load Disturbances

To evaluate the transient behavior of the controller, the feeder is subjected to time-varying disturbances caused by PV generation and load-demand variations. These disturbances modify the uncontrollable voltage component $\mathbf{v}^{\text{env}}$ in (1) and therefore create voltage deviations that must be corrected by inverter reactive-power control.

PV variability is represented through time-varying active-power injections at selected buses. These variations emulate the voltage effects of changing solar generation, including gradual generation changes and faster fluctuations. High PV penetration is known to introduce voltage-rise and voltage-regulation challenges in distribution feeders, especially when local reactive-power control is not coordinated across electrically coupled buses NRE (2025); Ste (2012); Turitsyn et al. (2011).

Load variability is modeled as time-varying demand at the non-slack buses. The load trajectories introduce additional changes in the voltage profile and create a nonstationary operating condition for the controller. The combined PV and load trajectories produce transient voltage deviations, allowing the comparison of recovery speed, control effort, and stability behavior under dynamic feeder conditions.

4.3. Controller Implementation

The evaluated controller consists of a graph/neighborhood-aware nominal reactive-power policy followed by a stability-oriented action filter. At each control step, each controllable inverter measures its local voltage and uses available one-hop controllable-neighbor voltage information to compute a nominal command. The nominal command is then scaled and constrained according to inverter command limits and ramp-rate limits before being dispatched.

The baseline controller corresponds to a local or graph-based safe controller without the same level of neighbor coordination. The proposed controller uses the neighborhood-aware graph potential described in Section III to reduce conflicting actions among controllable inverters. In both cases, the final reactive-power command is passed through the same safety-oriented filtering structure so that the comparison focuses on the benefit of incorporating neighbor information into the nominal control policy.

4.4. Performance Metrics

Controller performance is evaluated using three metrics that quantify transient recovery, accumulated transient performance, and stability-oriented behavior.

4.4.1. Mean Recovery Time

The mean recovery time measures how quickly the controller restores the feeder voltages to the admissible range after a disturbance. Let $\mathcal{T}_{\text{viol}}$ denote the set of disturbance events that cause at least one monitored bus voltage to violate the limits in (5). For each event, the recovery time is the elapsed time between the first voltage-limit violation and the first subsequent time instant at which all monitored bus voltages return to the admissible range. The reported mean recovery time is the average over all such disturbance events. A smaller value indicates faster transient voltage regulation.

4.4.2. Transient Cumulative Reward

The transient cumulative reward evaluates the overall transient performance during disturbance periods. It is computed by summing the instantaneous reward over the transient evaluation window. Equivalently, when the reward is defined as the negative of a voltage-regulation cost, a less negative cumulative reward indicates smaller voltage deviations and less aggressive control action. This metric captures both the duration and severity of voltage deviations during the transient response.

4.4.3. Mean Stability Index

In addition to recovery time and transient reward, the controller is evaluated using a stability-oriented index denoted by ΔF . This quantity measures the average change in the selected safety or Lyapunov-inspired function during closed-loop operation. A smaller value of ΔF indicates that the controller produces less growth in the stability-related quantity and therefore exhibits smoother closed-loop behavior. This metric is aligned with the stability-constrained control objective discussed in Section III and with prior Lyapunov-based approaches to safe reinforcement learning and voltage control Saf (2025); Lya (2024); Feng et al. (2024).

Together, these metrics evaluate the controller from three complementary perspectives: recovery speed, transient regulation quality, and stability-oriented safety behavior.

5. Case Study

5.1. Single-Phase IEEE-13-Inspired Distribution Feeder

The proposed graph/neighborhood-aware safe controller is evaluated on a single-phase radial distribution feeder inspired by the IEEE 13-node test system. The feeder is used as a representative benchmark for studying inverter-based Volt-VAR regulation under time-varying PV and load disturbances. The simulation considers three controllable inverter buses, located at buses 2, 7, and 9. These inverters provide reactive-power support, while the substation voltage is treated as fixed.

Fig. 2 shows the single-phase feeder model used in the case study. Since reactive-power injections at one bus can influence voltages at other electrically coupled buses, the feeder provides a suitable test case for evaluating neighborhood-aware inverter control. This coupling effect is a central issue in distributed Volt-VAR regulation and has been widely discussed in prior work on inverter-based voltage control Turitsyn et al. (2011); Dis (2015); Bolognani et al. (2019); Feng et al. (2024).

5.2. Voltage Response Under Dynamic Disturbances

Fig. 3 shows the voltage trajectories at the controllable buses under time-varying PV and load disturbances. The monitored voltages remain approximately within the admissible operating band during the disturbance period, with the lower envelope reaching about 0.95 p.u. and the upper envelope remaining below 1.00 p.u. Small numerical deviations at the voltage boundary are treated as simulation tolerance effects and do not change the observed recovery behavior.

The corresponding reactive-power commands are shown in Fig. 4. The controller adjusts inverter reactive-power injections in response to voltage deviations while respecting the imposed operating limits. This behavior is consistent with practical smart-inverter Volt-VAR regulation, where reactive-power support is used to mitigate voltage deviations caused by PV and load variability Ding et al. (2017); IEE (2020).

Fig. 5 shows the aggregated reactive-power response of the controllable inverter group. The aggregated curve represents the total reactive-power support required from the controlled buses during the disturbance sequence.

5.3. Performance Comparison

Table 2 summarizes the performance comparison between the reported baseline and the proposed graph/neighborhood-aware safe controller. The proposed controller reduces the mean recovery time from 2.6 s to 1.0 s, indicating faster

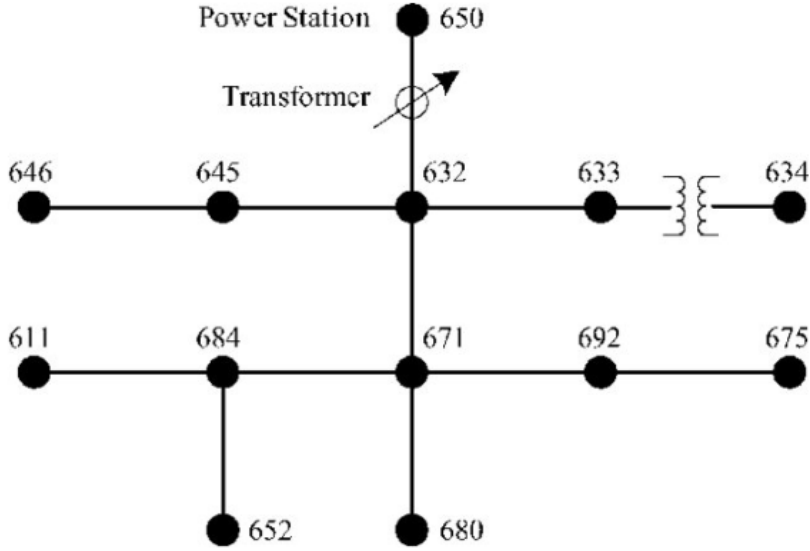


Fig. 2: Single-phase IEEE-13-inspired radial distribution feeder used for the case study.

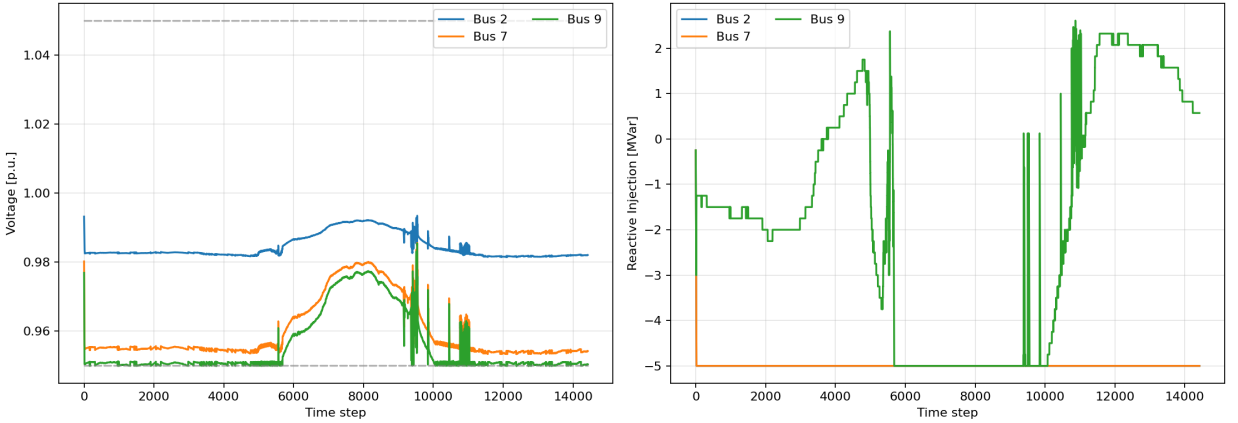


Fig. 3: Voltage profiles of the controllable buses under dynamic PV andload disturbances.

Table 2

Performance Comparison on the Single-Phase IEEE-13-Inspired Feeder

Metric	Baseline	Proposed
Mean recovery time (s)	2.6	1.0
Mean transient reward	-6.76	-1.50
Mean stability index, ΔF	-0.11	1.5×10^{-4}

restoration of the monitored voltages after disturbance events. The mean transient reward improves from -6.76 to -1.50 , showing smoother transient behavior and lower accumulated voltage-control penalty.

The third metric is reported as a stability-oriented index, denoted by ΔF , for the proposed controller. This value measures the average change in the selected stability-related function during closed-loop operation. A smaller magnitude indicates smoother stability behavior under the safety-filtered control action. Therefore, this metric should be interpreted as a stability-related evaluation index rather than a direct steady-state voltage violation value.

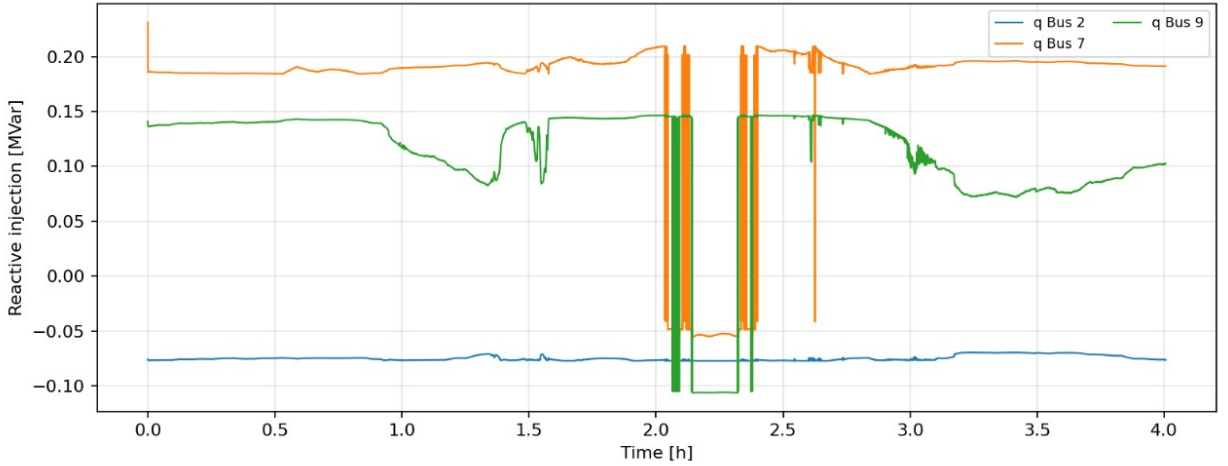


Fig. 4: Reactive-power commands at the controllable inverter buses.

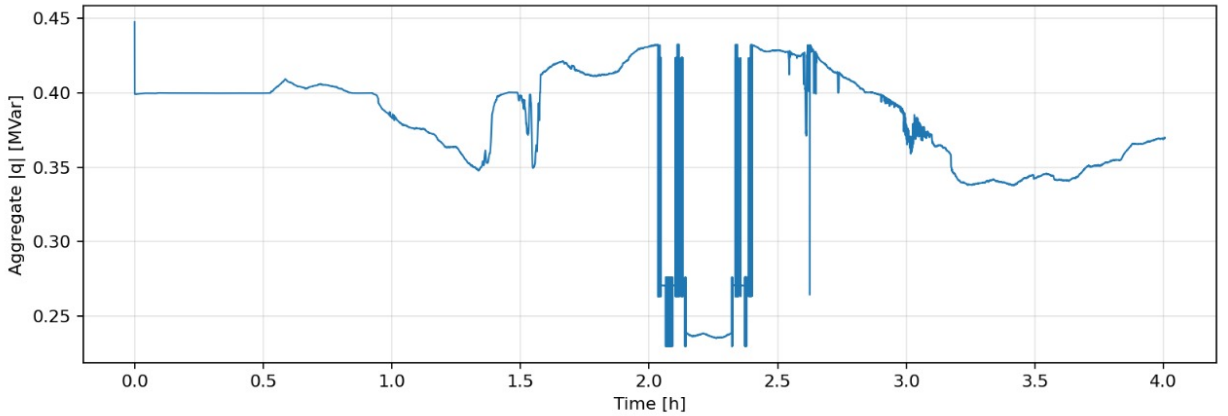


Fig. 5: Aggregated reactive-power response of the controllable inverters.

Overall, the results show that the proposed graph/neighborhood-aware safe controller improves transient recovery and reduces the accumulated transient penalty compared with the reported baseline. The stability-oriented filtering step keeps the dispatched inverter commands bounded by the prescribed control limits and avoids overly aggressive reactive-power changes. These results support the use of local neighborhood information and conservative safety filtering for real-time inverter-based voltage regulation in radial distribution feeders.

6. Conclusion

This work developed and evaluated a graph/neighborhood-aware safe Volt-VAR controller for inverter-interfaced distribution feeders under time-varying PV and load disturbances. The proposed controller uses local voltage feedback together with limited one-hop neighbor information to form a graph-based nominal reactive-power command, and then applies a stability-oriented action filter to enforce conservative gain scaling, inverter saturation limits, and ramp-rate constraints before dispatch. In the single-phase IEEE-13-inspired case study, the proposed method reduced the mean recovery time from 2.6 s to 1.0 s and improved the mean transient reward from -6.76 to -1.50 , while maintaining a small mean stability index of 1.5×10^{-4} . These results show that incorporating neighborhood information into the nominal control policy can improve transient voltage recovery and reduce accumulated voltage-control penalty without requiring a centralized optimal power flow solution or full feeder-wide observability. In real distribution systems with increasing PV penetration, such a controller can provide a lightweight and locally deployable option for fast inverter-based voltage regulation, helping utilities reduce voltage excursions and avoid conflicting reactive-power actions among nearby DERs. The main limitation of the present study is that the validation is performed on an equivalent single-phase radial feeder with a simplified neighborhood structure, so the results do not yet capture full three-phase unbalance, detailed inverter dynamics, communication delay, measurement noise, or feeder reconfiguration during faults. Future work will extend the controller to full three-phase unbalanced feeders, larger and reconfigurable distribution networks, delay-aware neighbor communication, hardware-in-the-loop testing, and adaptive tuning of the stability filter under changing operating conditions.

Declaration of Generative AI and AI-assisted Technologies in the Writing Process

During the preparation of this work the author(s) used Grammarly in order to proofread the manuscript and improve the clarity of the English writing. After using this tool/service, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the published article.

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